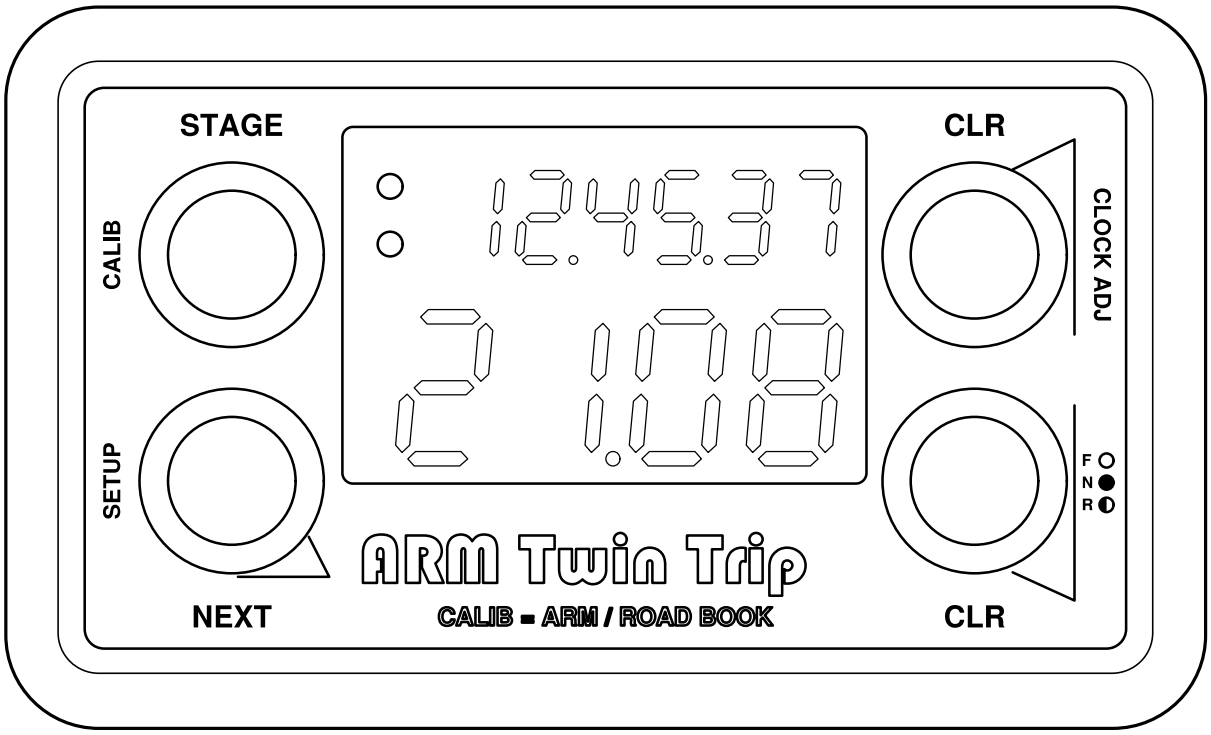
ARM products / Pulse Trip Meter

ARM Twin Trip

SS2027

User Manual

Please read this manual before use and keep it for future reference.



Contents

01	Safety precautions
02	Parts
03	Power and clock
04	Switching the upper display
05	Trips (Trip A / Trip B)
06	Distance calibration
07	Speed and maximum speed
08	Speed alert
09	Stage timing and stage records
10	Ave and final time
11	Settings (brightness, pulses, unit, Trip B display)
12	Remote and external switches
13	Driver display (optional)
14	Rear panel (mounting, connector)
15	Specifications
16	Quick reference

CHAPTER 01

Safety precautions

▲ Power supply
Power is taken from DC+12V and GND on the rear connector. The unit cannot be powered by USB. Wrong wiring may damage the unit.

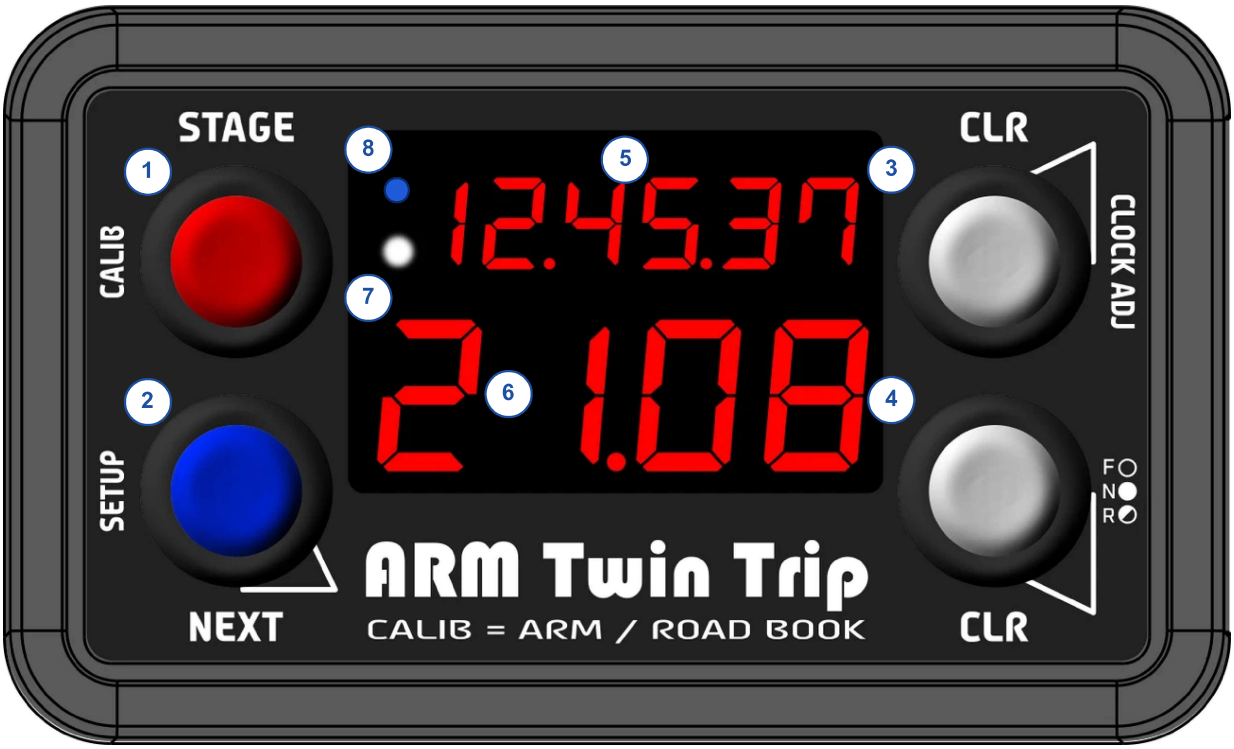
▲ Operating while driving
All operation while driving should be done by the co-driver. The driver must concentrate on driving; mount the driver display where it can be read with minimal eye movement.

How distance is measured
Distance and speed are measured from the vehicle speed pulse (SPD). Set the number of pulses and calibrate the distance before use.

Mounting location
Do not use or store the unit where it can get wet or hot (e.g. in direct sunlight on the dashboard).

CHAPTER 02

Parts



No.	Name	Function
1	Red button (STAGE / CALIB)	Short press: stage timing. Long press on the clock screen: distance calibration.
2	Blue button (NEXT / SETUP)	Short press: switches the upper display. Long press: settings (brightness, pulses, unit, Trip B display).
3	Upper right CLR (CLOCK ADJ)	Acts on the upper display (reset etc.). Long press on the clock screen: clock adjust. Increases the value while setting.
4	Lower right CLR (F / N / R)	Short press: resets Trip B. Long press: switches the direction. Decreases the value while setting.
5	Upper display	Clock, Trip A, speed, final time or stage records.
6	Lower display	Trip B (section distance).
7	White LED (direction)	Off: forward (F). On: stop (N). Blinking: reverse (R).
8	Blue LED	Final time warning (chapter 10).

Side: 3.5 mm jack for the remote (Trip B reset). Right side: power switch at the top (rocker switch: | ON / ○ OFF) and driver display connector at the bottom (Molex Mini 6-pin).

Rear: 6-pin connector for power, speed pulse and external switches, and mounting insert nuts (chapter 14).

CHAPTER 03

Power and clock

Power

1. Connect DC+12V and GND of the rear connector to the vehicle.
2. The rocker switch at the top of the right side is the power switch. Press the | side for ON and the ○ side for OFF.
3. The clock runs on the built-in real-time clock (RTC) and keeps running on its backup battery while the power is off.

Clock adjust

1. On the clock screen, long press the upper right CLR.
2. Upper right CLR adds 1 to the blinking digit (hours, minutes, seconds), lower right CLR subtracts 1. The blue button moves to the next digit.
3. Press the red button to confirm. To leave without changing, long press the upper right CLR.

Clock battery

If the clock stopped because its backup battery ran out, “bAt Lo” is shown on the upper display for 2 seconds at power-on, and the whole time blinks on the clock screen. Meanwhile the final time screen shows “Ft rtc”. Set the clock to return to normal.



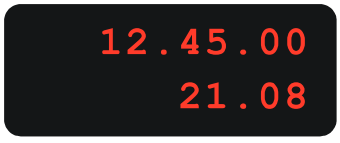
At power-on (2 s)

Time hold

On the clock screen, a short press of the upper right CLR keeps showing the time at the moment you pressed, blinking. Press again to return.



Clock



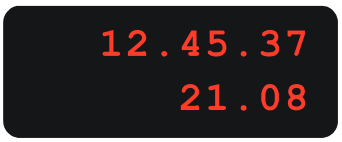
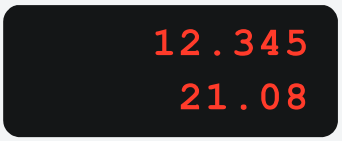
Clock adjust (seconds)

What is kept
Trip A / Trip B distances (saved every 50 m), calibration, pulses, unit, Trip B display format, brightness, speed limit and Ave settings are kept when the power is off. Up to the last 50 m driven before switching off may not be restored. Stage records are lost when the power is off.

CHAPTER 04

Switching the upper display

Each short press of the blue button switches the upper display as below. The lower display normally always shows Trip B.

Screen	Upper display	Example
Clock	h.mm.ss	
Trip A	Total distance (km)	
Speed	“S” followed by the calibrated speed (km/h)	
Final time	Ahead / behind against the Ave (chapter 10)	
Stage records	Stage time. The lower display shows the record number	

After stage records, the display returns to the clock.

CHAPTER 05

Trips (Trip A / Trip B)

Resetting the trips

On the Trip A screen, a short press of the upper right CLR resets Trip A, and a short press of the lower right CLR resets Trip B. Trip B can also be reset on the clock, speed and final time screens and during a stage.

Trip A display unit

With the “Unit” setting, Trip A is shown in 1 m steps (12.345) or 10 m steps (12.34) (chapter 11).

Direction

Each long press of the lower right CLR switches forward (F) → stop (N) → reverse (R) → forward (F).

Direction	Action	White LED
F	Distance increases.	Off
N	Distance stops.	On
R	Distance decreases (e.g. to recover after a wrong turn).	Blinking

It works on Trip A, Trip B and Trip D on the driver display.

CHAPTER 06

Distance calibration

When the displayed distance does not match the road book, adjust it with the calibration factor.

Example: a section of 10.00 km in the road book shows 10.20 km on Trip A.

1. On the clock screen, long press the red button to enter calibration (C).
2. Upper right CLR adds 1 to the blinking digit, lower right CLR subtracts 1. The blue button moves to the next digit.
3. Set 10.20 ÷ 10.00 = 1.0200.
4. A short press of the red button saves and returns to the clock.

How to calculate
Calibration factor = ARM displayed distance ÷ road book distance



Calibration screen



Set to 1.0200

A long press of the red button during calibration returns the factor to 1.0000. The factor also applies to the speed display. Set the number of pulses for your vehicle first (chapter 11).

CHAPTER 07

Speed and maximum speed

The speed screen shows “S” followed by the calibrated speed. A short press of the upper right CLR shows the maximum speed (P); press again to return. The maximum speed resets to 0 when a stage starts.



Speed



Maximum speed

CHAPTER 08

Speed alert

1. On the speed screen, long press the upper right CLR to open the speed alert setting (A).
2. Upper right CLR adds 10, lower right CLR subtracts 10 (10–90 km/h).
3. The red button turns the alert ON and returns to the speed screen, which then shows “A”.
4. To turn it OFF, open the setting and press the blue button.

Speed	Upper display / white LED on the driver display
Up to 5 km/h below the limit	Normal display
From 5 km/h below the limit	Blinks slowly
At or above the limit	Blinks fast



Setting screen



Speed with the alert ON

CHAPTER 09

Stage timing and stage records

Stage timing

1. A short press of the red button starts a countdown to 00 s of the next minute. Trip A and Trip B are reset to 0 automatically.
2. If the car starts moving during the countdown (3 speed pulses), the stage starts at that moment. If it does not move, the stage starts at 00 s. You can also press the red button during the countdown to start immediately.
3. While running, the upper display shows the stage time. The blue button switches time ⇔ Trip A (the “S” on the left blinks while Trip A is shown).
4. Every 1 km, the passing time blinks for 3 seconds.
5. Press the red button at the finish. The time blinks and Trip B returns to 0.
6. Press the red button again to end the stage. The clock screen shows the finish time blinking (clear it with the upper right CLR or the red button).



Countdown | Running | Finish

The blue button cancels during the countdown. On the stage screen, the upper right CLR leaves the stage screen. The external switch (stage) does the same as the red button.

Stage records

Switch the upper display to “stage records” to view the recorded times. The lower display shows the record number (St01–St30). Upper right CLR shows the next record, lower right CLR the previous one. Up to 30 records are kept; they are lost when the power is off.

CHAPTER 10

Ave and final time

When an Ave (target average speed) is set, the final time shows the difference from driving at the Ave, based on the distance and elapsed time since the stage start. When the Ave is OFF, “OFF” is shown.

This function gives an estimate of the time left until your TC in time when you arrive at the next time control (TC). It helps you check how much spare time you have.

How to set

1. On the final time screen, a short press of the upper right CLR opens the Ave setting.
2. Set the Ave on the lower display with the upper right CLR: OFF → 30 → 40 … The blue button moves to the next digit.
3. The upper display shows the start time. It is set automatically to the start time when a stage starts.
4. Long press the upper right CLR to save (the distance so far is kept). Saving with the red button restarts the distance from that point.



21 s ahead



1 min 5 s behind

Blue LED warning
The blue LED (above the white LED) shows the final time margin: it blinks fast when the margin is under 5 minutes (including when behind), slowly from 5 to 10 minutes, and is off otherwise. It works once an Ave is set and a stage has been started (off while the stage screen is shown).

CHAPTER 11

Settings (brightness, pulses, unit, Trip B display)

Long press the blue button to enter the settings; each short press of the blue button moves to the next setting. The last press returns to the previous screen.

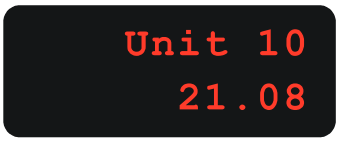
Order	Display	Setting	Operation
1	LEd	Brightness (3 levels, 0 to 2).	Upper right: brighter / lower right: dimmer
2	PULS	Speed pulses (2, 4, 8). Set to match the vehicle.	Upper / lower right: switch
3	Unit	Trip A unit. “1” is 1 m (12.345), “10” is 10 m (12.34).	Upper / lower right: switch
4	tr-	Trip B display. “0.01” is normal (12.34 km); “10” enlarges distances under 10 km (1.230 km).	Upper / lower right: switch



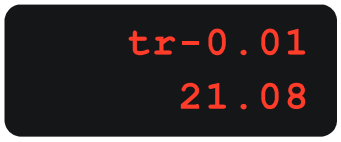
Brightness



Pulses



Unit



Trip B display

The settings are kept when the power is off.

About the Trip B distance format

The “10” mode is mainly for recce: it lets you judge the length of straights and corners in 10 m steps. Seeing 10 m, 20 m … makes it easier to write clear pace notes.

CHAPTER 12

Remote and external switches

Remote (3.5 mm jack)

Connect the supplied remote to the 3.5 mm jack on the side to reset Trip B from your hand.

Button	Function
Remote (supplied)	Resets Trip B

Mount it near the steering wheel or gear lever to reset Trip B at each road book box without reaching for the unit.

External switches (rear connector)

Wire	Function
White (Trip B reset)	Trip B reset only. A short press resets Trip B. Useful for adding a foot switch.
Yellow (Stage)	Adds only the red button's stage function (start / finish). A long press does nothing (it does not open calibration).

Wiring
Connect one side of a momentary switch (ON only while pressed) to the white or yellow wire and the other side to the vehicle GND. If the vehicle GND is not available, use the GND on pin 6 of the connector.

CHAPTER 13

Driver display (optional)

The driver display is an optional accessory.

Connect a driver display to the connector on the lower right side (Molex Mini 6-pin) to show a dedicated trip (Trip D) in front of the driver.

Item	Details
Display	Trip D (km). Under 10 km it is enlarged, e.g. 1.230 km.
Reset	The reset button on the driver display resets Trip D. It also returns to 0 at the stage start and finish.
White LED	Lights in step with the blinking upper display while the speed alert is ON.
Direction	Counts according to F / N / R, the same as the main unit.



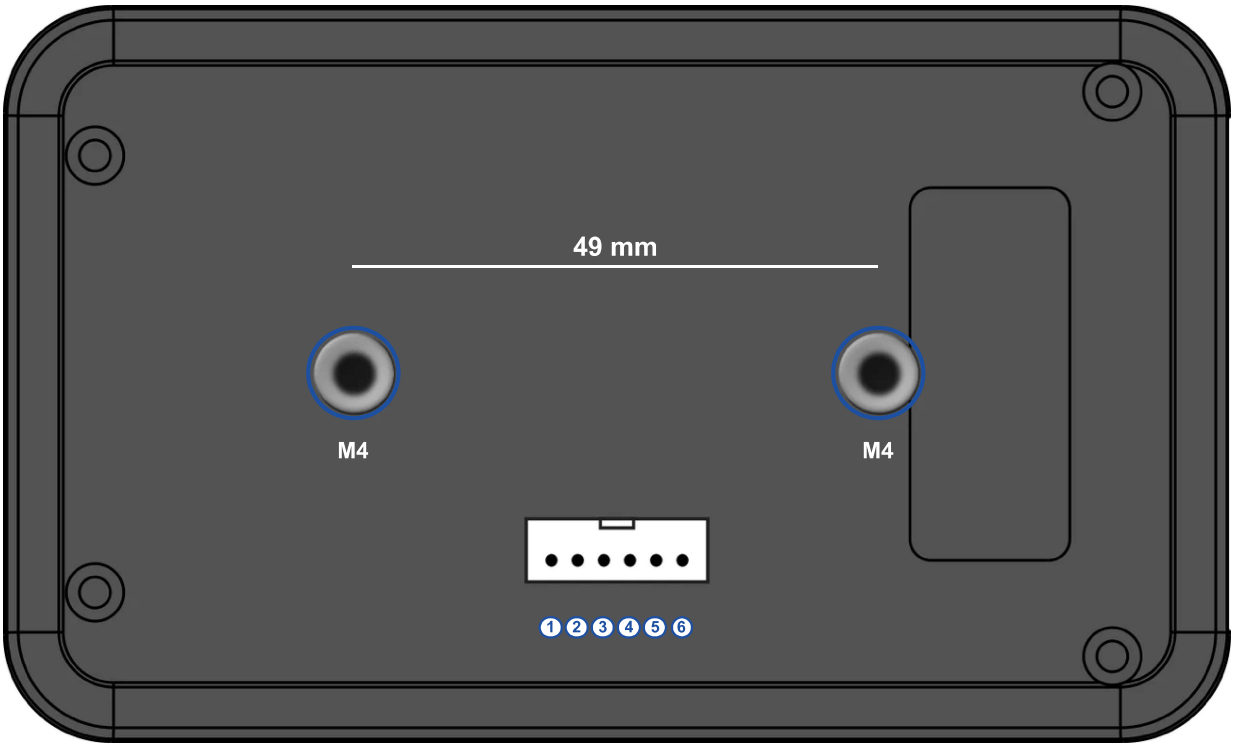
Trip D (under 10 km)



Trip D (10 km or more)

CHAPTER 14

Rear panel (mounting, connector)



Mounting insert nuts

Two mounting nuts are embedded in the rear, so the unit can be bolted to a bracket or mount.

Item	Details
Bolt	M4
Bolt length	Up to the thickness of the mounting part (bracket etc.) + 10 mm
Nut spacing	49 mm (centre to centre). RAM Mount bases with the same spacing can be used.

▲ Caution
A bolt that is too long may damage the inside of the unit. Always use bolts no longer than “thickness + 10 mm”.

Connector (XA 6-pin)

Pin (from the left, seen from the rear)	Use
1	DC+12V (power)
2	GND
3	SPD (vehicle speed pulse input)
4	White: external switch (Trip B reset)
5	Yellow: external switch (stage)
6	GND (also usable for the external switches)

> **▲ Caution**
> Power is taken only from this connector. Take care not to swap DC+12V and GND.

CHAPTER 15

Specifications

Item	Details
Dimensions	W 115 × H 69 × D 28 mm (excluding buttons and other protrusions)
Power	DC12V (rear connector), power switch on the right side. No USB power.
Distance / speed	Measured from the vehicle speed pulse. Pulses 2, 4 or 8; calibration factor 0.0000–2.9999
Clock	Built-in real-time clock (RTC) with backup battery; a dead battery is shown on the display
Memory	Trip A / Trip B distances (every 50 m), calibration, pulses, unit, Trip B display format, brightness, speed limit, Ave settings (stage records are lost when the power is off)
Connections	Rear XA 6-pin connector (power, speed pulse, external switches), remote (3.5 mm on the side), driver display (Molex Mini 6-pin, lower right side)
Supplied accessory	Remote (Trip B reset)
Options	Driver display

CHAPTER 16

Quick reference

Screen	Blue	Red	Upper right CLR	Lower right CLR
Clock	Short: next screen Long: settings	Short: stage Long: calibration	Short: time hold Long: clock adjust	Short: Trip B reset Long: direction
Trip A	Short: next screen	Short: stage	Short: Trip A reset	Short: Trip B reset Long: direction
Speed	Short: next screen	Short: stage	Short: speed ⇔ maximum Long: speed alert setting	Short: Trip B reset Long: direction
Final time	Short: next screen	Short: stage	Short: Ave setting Long: stop the display	Short: Trip B reset Long: direction
Stage records	Short: to clock	Short: stage	Next record	Previous record
Stage	Countdown: cancel Running: time ⇔ Trip A	Countdown → start → finish → end	Leave the stage screen	Trip B reset

Setting screen	Upper right	Lower right	Blue	Confirm / back
Calibration (C)	+1	−1	Next digit	Red short: save / Red long: 1.0000
Clock adjust	+1	−1	Next digit	Red: confirm / Upper right long: back
LEd / PULS / Unit / tr-	+ / switch	− / switch	Next setting	Last blue press returns
Speed alert (A)	+10	−10	Back with alert OFF	Red: back with alert ON
Ave setting	+	−	Next digit	Upper right long: save / Red: save and restart the distance

An animated manual is also available: manual.rally-arm.com